Calcurion Maths Worksheets

A-Level Mathematics

Topic: Pure 1 – Quadratics

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Solve $x^2 - 5x + 6 = 0$. **[3]**

2. Solve $x^2 + 7x + 12 = 0$. **[3]**

3. Solve $2x^2 - 5x - 3 = 0$. **[4]**

4. Solve $3x^2 + x - 2 = 0$. **[4]**

5. Complete the square: $x^2 + 6x + 5$. **[3]**

6. Complete the square: $x^2 - 8x + 3$. **[3]**

7. Complete the square: $2x^2 + 8x - 1$. **[4]**

8. Find the minimum value of $y = x^2 - 4x + 7$. **[3]**

9. Find the maximum value of $y = 5 - 2x - x^2$. **[4]**

10. Solve $x^2 - 4x - 1 = 0$ giving exact answers. **[3]**

11. Solve $2x^2 + 3x - 1 = 0$ giving exact answers. **[4]**

12. Sketch $y = (x - 2)(x + 3)$, showing intercepts. **[4]**

13. Sketch $y = x^2 - 2x - 8$, showing intercepts and vertex. **[5]**

14. Find the discriminant of $3x^2 - 5x + 2 = 0$ and state the nature of roots. **[3]**

15. Find the discriminant of $x^2 + 2x + 5 = 0$ and state the nature of roots. **[3]**

Section B – Hard (Q16–25)

16. Find the range of values of k for which $x^2 + kx + 4 = 0$ has real roots. **[4]**

17. Find the range of values of k for which $kx^2 + 3x + k = 0$ has no real roots. **[5]**

18. Solve $x^4 - 5x^2 + 4 = 0$. **[5]**

19. Solve $x^4 - 13x^2 + 36 = 0$. **[5]**

20. The equation $x^2 - 2x + k = 0$ has equal roots. Find k. **[3]**

21. By completing the square, solve $x^2 + 10x + 18 = 0$. **[4]**

22. Find the coordinates of the turning point of $y = 2x^2 - 12x + 7$. **[4]**

23. Solve the simultaneous equations $y = x^2 - 3x + 2$ and $y = 2x - 1$. **[5]**

24. Solve the simultaneous equations $y = x^2 + x$ and $y = 3x + 2$. **[5]**

25. Prove that $x^2 + 4x + 5 > 0$ for all real x. **[4]**

End of Worksheet • *Show all working clearly.*